

Suggested Mark Scheme

- 1 Fig. 1.1 shows a structure found in an animal cell that is crucial in the cell cycle.

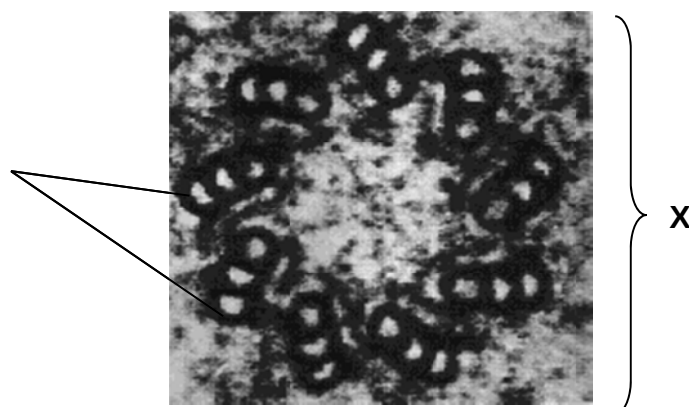


Fig. 1.1

- (a)(i) Name structure X.

centriole;

[1]

- (ii) Describe arrangement of structure X.

1. 9 sets of triplet microtubules;

2. arranged in a ring-like struc forming hollow cylinder;

3. exists in a pair (a pair of centrioles);

4. at right angles to each other;

[2]

- (iii) Explain the role of structure X in the M phase of cell cycle.

1. part of microtubules organising centre (MTOC);

2. microtubules radiate from centrosome containing pair of centrioles;

3. microtubules elongate, attach to kinetochore on centromere of sis chromatids during metaphase;

4. contraction & shortening of microtubules pull sis chromatids to opp poles of cell;

5. results in equal separation of sis chromatids during anaphase;

[2]

- (b) Describe the basic unit of structure Y.

1. tubulin made of aa connected by peptide bds forming linear polypeptide;

2. folded to form a tertiary struc (3D config) with bonds b/w R gps of aas;

3. 2 polypeptide chains interact to form a dimer with 4^o struc of prot;

4. with interxns/bonds b/w two 3^o struc of prots;

[2]

Changes in DNA amount and chromosome numbers were monitored during one cycle of meiosis. Results are shown in Fig. 1.2.

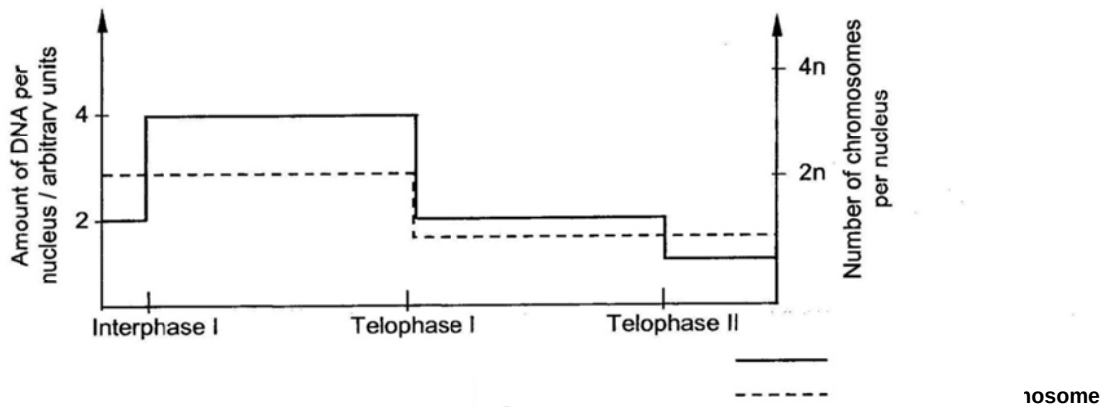


Fig. 1.2

(c) Explain why DNA amount and chromosome numbers remain constant during the two anaphases in Fig. 1.2.

1. DNA amt & chr no. is with ref to a nucleus within 1 cell;

2. during anaphase I & II, homologous chr & sis chromatids are moving towards pole respectively;

3. nucleus is not formed yet, cell has not divided into 2 indiv cells;

4. content & chr no. will remain unchanged/will remain const;

[2]

(d) Explain the significance of meiosis in genetic variation.

1. prophase I, crossing over b/w non-sis chromatids of homologous chr forming chiasmata;

2. exchange of segments of chr result in new combination of alleles in gametes formed;

3. metaphase I & metaphase II, independent assortment/arrangement of homologous chr & sis chromatids at metaphase plate;

4. gives rise to diff combinations of chromosomes in gametes;

5. anaphase I & anaphase II, random segregation of homologous chr & sis chromatids to opp poles of cell;

6. results in gamete cells having diff combination of alleles of chr;

[3]

[Total: 12]

- 2 Fig. 2 shows the sequence of bases in a section of the DNA template strand coding for the first seven amino acids of a polypeptide chain.

3' TACAAGGTCGTCTTTGTCAA 5'

Fig. 2

- (a)(i) List the corresponding codons in the mRNA transcribed from the template in Fig. 2.

AUG UUC CAG CAG AAA CAG UUU [1]

- (ii) Methionine is coded by the initiation codon. On Fig. 2, label, the 3' and 5' end of the DNA template strand. [1]

The polypeptide was hydrolysed to release the first seven amino acids. It contained four different amino acids. The number of each type obtained is shown in the Table 2.

Table 2

Amino acid	Number present
Phe	2
Met	1
Lys	1
Gln	3

- (b) Use the base sequence shown in Fig. 2 to work out the order of amino acids in the polypeptide. [1]

met phe gln gln lys gln phe

- (c)(i) Explain why single base deletions in the DNA sequence is more likely to produce non-functional proteins than single base substitutions.

1. del cause frameshift;

2. all aas downstream of mutation Δ ; ® nonsense mutation

3. sub usually cause missense / Δ in 1 aa;

4. or silent mutation if sub occurs at 3rd base as mutation leads to degenerate codes due to wobble base phenomenon;

5. Δ in aa seq more drastic / more R grps / folding Δ in del, thus prot more likely to have vastly diff 3D config leading to lost in fn; [2]

- (ii) Describe the effect of a transition mutation at base position 6.

1. G $\rightarrow$ A sub

2. AAG $\rightarrow$ AAA / UUC $\rightarrow$ UUU

3. results in degenerate code;

4. *still code for phe / no Δ in aa, silent mutation;* [2]

(d) Explain how the structure of the ribosome is suited to perform its function.

1. *small s.u. with recog / binding site for initiation tRNA;*

2. *binding site for 5' end of mRNA to initiate translation;*

3. *large s.u. with P site to bind tRNA with growing polypep chain;*

4. *with A site to bind incoming tRNA ;*

5. *with E site to allow tRNA exit;*

6. *with peptidyl transferase to catalyse peptide bond;*

7. *binding site for translation factors / GTP / release factor etc.* [3]

[Total: 10]

3 Fig. 3.1 shows a bacterium undergoing the process of binary fission to produce clones.

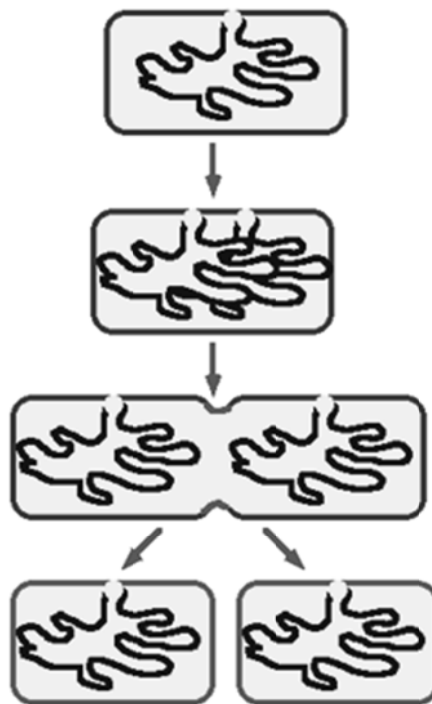


Fig. 3.1

(a) Explain why these newly synthesized are described as *clones*.

1. *genetically identical to each other & parent cell;*

2. *have same alleles at all gene loci & originate from same parent;* [1]

(b) Describe how binary fission in bacteria brings about genetic stability.

1. *duplication of chromosome occurs;*

2. DNA is replicated via semi-conservative replication;

3. genetic material divided equally to 2 daughter cells as elongated cell divides into two;

4. ensure genetically identical daughter cells from 1 parent cell; [2]

While binary fission ensures genetic stability, genetic variation can still occur in bacteria through genetic exchange and recombination. A process that brings about variation is conjugation.

(c) Describe how genetic variation is brought about through conjugation.

1. F^+ cell with F factor forms a sex pilus/mating bridge (physical contact) with F cell;

2. DNA/ F factor transferred from F^+ to F cell via rolling circle mechanism;

3. exchange of alleles at homologous regions on F chr via homologous recombination;

4. result in F acquiring alleles from F^+ cell leading to genetic variation; [2]

Fig. 3.2 shows a molecular structure of a *lac* repressor that regulates the activity of *lac* operon in a bacterium such as *Escherichia coli*.

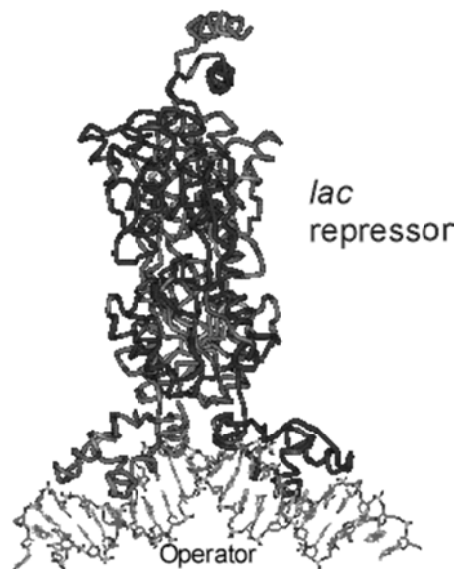


Fig. 3.2

A strain of the bacterium, *Escherichia coli*, has been discovered that has a mutation in the regulator gene. One aspect of the phenotype of this mutant is that it produces large quantities of β -galactosidase at all times.

(d) Explain why a mutation in the regulator gene leads to the constant production of β -galactosidase.

- 1. change in base seq of regulator gene;**
- 2. no pdtn of repressor prot/ pdc repressor with diff 3D config;**
- 3. no repressor binds to operator region of lac operon even in absence of lactose/ binding site of repressor not complementary to operator region;**
- 4. RNA pol continuously binds to promoter;**
- 5. RNA pol activity unblocked;**
- 6. continuous transcription of lac Z gene to pdc β -galactosidase prot;**

[3]

A certain bacteriophage is known to insert its own genome between the *lacZ* and *lacY* structural genes to give prophages.

- (e) Explain the change in number of bacteriophage (if any) in conditions when host bacteria are grown in environment with

- (i) abundant supply of glucose and no lactose supply;

- 1. no allolactose binds to allosteric site of repressor;**
- 2. repressor remains bound to operator region, operon not switched on;**
- 3. no transcription of structural genes & viral genes by RNA pol;**
- 4. no pdtn of phage prots, thus no new phage particles formed;**

[2]

- (ii) no supply of glucose and abundant supply of lactose.

- 1. allolactose binds to allosteric site of repressor;**
- 2. repressor dissociates from operator, operon switched on;**
- 3. transcription of structural genes & viral genes by RNA pol;**
- 4. pdtn of phage prots, new phage particles formed;**

[2]

[Total: 12]

- 4 Cancer cells exhibit a set of unique properties that distinguish them from their normal counterparts. Cancer cells frequently exhibit malfunction in cell cycle checkpoints. One such example is the regulation of passage from G_1 to S phase.

A series of regulatory interactions involving E2F family of transcription factors and Rb proteins are involved in the G_1 to S phase progression. E2F up-regulates transcription of proto-oncogenes whose products are required for passage from G_1 to S phase. The Rb protein represses E2F activity through protein-protein interactions.

The binding of Rb protein to E2F is regulated by Rb's phosphorylation status.

Hypophosphorylated Rb binds to E2F while hyperphosphorylated Rb is released from this interaction.

- (a) State **two** properties of cancer cells that distinguish them from their normal counterparts.

1. **active telomerase / immortal cell line; ® immortal cells**
2. **bypass DNA damage / DNA replication / spindle chkpts;**
3. **uncontrolled cell proliferation / avoids apoptosis / G1 arrest;**
4. **unspecialised;**
5. **⊗ anchorage dependence / contact inhibition**

[2]

- (b) Explain how a mutation in the Rb gene affecting the phosphorylation status of Rb can result in the development of cancer.

1. **pt mutation in Rb gene => Δ in aa seq / 3D config of RB prot;**
2. **result hyperphosphorylation / not dephosphorylated by phosphatase;**
3. **⊗ bind to E2F;**
4. **E2F is active & available in binding to reg seq of proto-oncogenes;**
5. **lead to gain-in-fn mutation of proto-oncogene, becomes oncogene;**
6. **↑ pdtn of prot that ⊕ cell div leading to uncontrolled cell proliferation / div in abs of GF thus tumour formation;**

[3]

- (c) Some cancer cells also exhibit mutations in genes coding for histone deacetylase. Suggest how such mutations may lead to cancerous growth.

1. **pt mutations in gene coding for deacetylase => Δ in aa seq thus prot conformation;**
2. **results in enz gain-in-fn / **loss-in-fn**;**
3. **removes / **unable to remove** acetyl grps fr histones;**
4. **↑ / ↓ packing of chromatin containing TSG / **proto-oncogenes**, thus silencing / **unable to repress exp**;**

[2]

- (d) With reference to transcriptional control, describe how the influence of E2F on the expression of proto-oncogenes can be balanced in a eukaryotic cell.

1. **repressors bound to silencers**
2. **interxt with RNA pol-TF cplx (via prot-prot interxns at repression domain) to reduce stability of TIC;**
3. **repressors bind to activation / DNA-binding domain of E2F;**
4. **to ⊗ interxn with TFs / enhancer;**
5. **repressors bind to silencer seq which overlaps with enhancer;**
6. **compete for /⊗ E2F from binding to enhancer;**

[3]

- (e) Explain why cancer usually occurs in older individuals.

1. **cancer is a multi-step process;**
2. **devt req mutations in many genes in the same cell;**
3. **thus req time to accumulate;**

4. *older indivls exposed to mutagens over lifetime, ting chance of mutations;* [2]

[Total: 12]

5 (a) Explain what is meant by *homologous pairs of chromosomes*.

1. *chr with same length, same position of centromere;*

2. *same banding pattern, same genes & gene loci;*

3. *diff alleles of same gene;*

4. *from diff origin (1 from male parent, 1 from female parent);* [2]

An inbred variety of maize, **A**, with finely striped leaves was found to be resistant to the fungus that causes the disease, corn leaf blight.

Plants of variety **A** were crossed with another inbred variety of maize, **B**, which had entirely green leaves and low resistance to the fungus. All the F_1 generation had entirely green leaves and low resistance.

(b)(i) Using appropriate symbols and indicating what they represent, state the genotype of **A**.

1. *G - entirely green leaf, R – low resistance;*

2. *genotype of A: ggrr;* [1]

(ii) The above F_1 generation was back-crossed to variety **A**. Using a genetic diagram, show the expected phenotype of the F_2 generation from this cross.

Phenotype *entirely green, low resistance* x *striped resistant leaves,* [1]

[Genotype *GgRr* x *ggrr* [2]

Meiosis

Gametes *GR* *Gr* *gR* *gr* *gr* [3]

Fertilisation

Using Punnett Square

	<i>GR</i>	<i>gR</i>	<i>Gr</i>	<i>gr</i>
<i>gr</i>	<i>GgRr</i>	<i>ggRr</i>	<i>Ggrr</i>	<i>ggrr</i>

F_2 genotype *GgRr* *ggRr* *Ggrr* *ggrr* [4]

F_2 phenotype *entirely green, low* *striped leaves, low* *entirely green,* *striped leaves,*

(offspring) resistance resistance resistant resistant [5]
*F*₂ phenotypic 1 : 1 : 1 : 1 [6]
 ratio

[3]

The above was carried out (*F*₁ generation back-crossed to variety **A**) and yielded the following results:

finely striped leaves and resistance	80
finely striped leaves and low resistance	20
entirely green leaves and resistance	22
entirely green leaves and low resistance	78

The χ^2 (chi square) test was done on these data, giving a calculated value of $\chi^2 = 67.36$.

degree of freedom	probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.31	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

Table 5.1

- (c)(i) Use the calculated value of χ^2 and the table of probabilities provided in Table 5.1 to find the probability of the results of the cross departing significantly by chance from the expected ratio.

Probability: **$p < 0.001$** [1]

- (ii) State what conclusions may be drawn from the probability found in (c)(i).

1. since χ^2 calculated > χ^2 critical; null hypothesis is rejected;

2. there is a sig diff^{ce} b/w O & E nos. of phenotypic results;

3. distribution does not follow expected 1:1:1:1 ratio;

4. occurrence of diff^{ce} is not due to chance;

[2]

- (iii) Explain the discrepancy between the actual and expected results.

1. 2 genes, leaf colour/pattern & resistance are linked;

2. genes are found on same chr;

3. *do not assort & segregate independently at anaphase of meiosis*

4. *parental phenotypes (finely striped leaves + resistant, entirely green leaves + low resistance) in larger proportions;*

5. *recombinant phenotypes (finely striped leaves + low resistance, entirely green leaves + resistant) arise due to crossing over during prophase 1 of meiosis;*

6. *due to lower chances of occurrence for crossing over, proportions lower for recombinants than parental phenotypes;* [3]

[Total: 12]

- 6 Fig.6.1 shows how a stimulating electrode was used to change the potential difference across an axon membrane. Two other electrodes, **P** and **Q**, were used to record any potential difference produced after stimulation.

The experiment was repeated six times, using a different stimulus potential each time. In experiments **1** to **4**, the stimulating voltage made the inside of the axon less negative. In experiments **5** and **6**, it made the inside of the axon more negative.

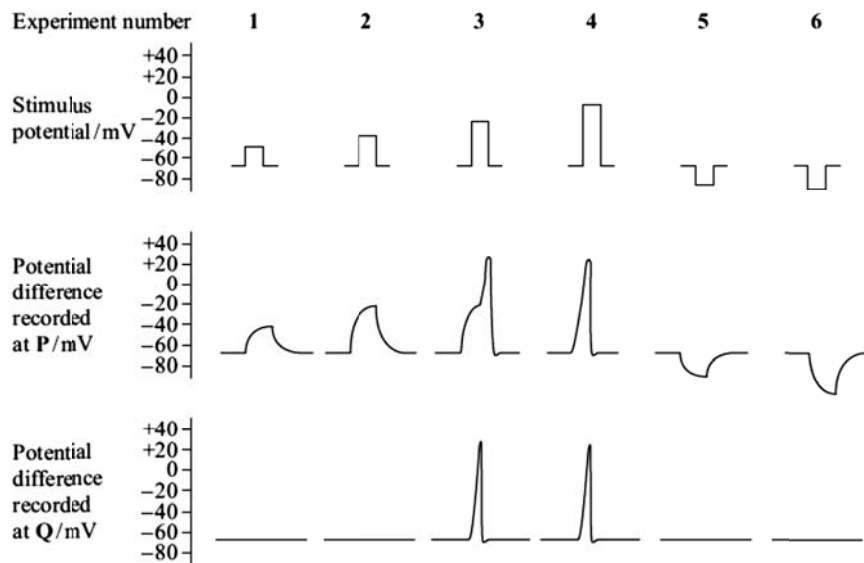


Fig. 6.1

- (a) Explain the results of experiments 2 and 3.

- (i) 2
1. *increase stimulus pot from -70 to -40mV, causes rise in pot diff at P from -70 to -20mV;*
 2. *depolarization at P pdces a graded potential;*
 3. *below threshold pot, no A.P. generated;*
 4. *no impulse/A.P. propagated/transmitted to Q;*

[2]

- (ii) 3
1. *further increase in stimulus pot from -70 to -20mV, causes further rise in pot diff at P from -70 to -20mV;*
 2. *depolarization at P exceeds threshold pot, from -70mV to +30mV;*
 3. *opening of voltage gated Na^+ channels in axon memb, cause influx of Na^+ into axoplasm;*
 4. *A.P. generated at P & propagated/transmitted to Q;*
- [2]

Myasthenia gravis is a disease which causes muscular weakness. It develops because of an attack by the body's own immune system on neuromuscular junctions. Fig.6.2 below shows a normal neuromuscular junction (nerve to muscle) and one affected by the disease (myasthenic).

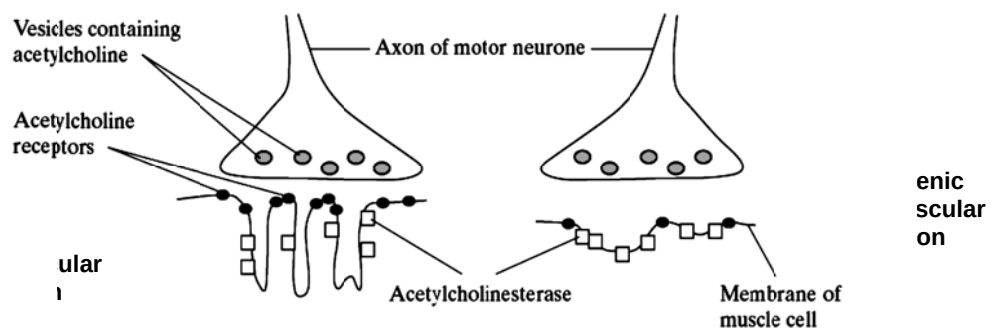


Fig. 6.2

- (b) Describe **two** ways in which a myasthenic neuromuscular junction differs from a normal one and explain how each difference would affect transmissions across the myasthenic neuromuscular junction.

1. *myasthenic has fewer folds, less SA, thus fewer ACh receptors exposed;*
- (i) Difference 1:
2. *less chance of Ach to bind to receptors on post-synaptic memb;*
- Effect:
3. *less ligand gated Na^+ channels present thus slower Na^+ influx when open;*
 4. *less chance for depolarization/ EPSP generation to occur;*
- [2]
5. *myasthenic has more acetylcholinesterase exposed on post-synaptic memb/lower ratio of receptors to acetylcholinesterase;*
- (ii) Difference 2:
- Effect:
6. *higher chance Ach degraded before binding to Ach receptors;*
 7. *less chance of binding of Ach to receptors on post-synaptic memb;*
 8. *less chance of depolarization/EPSP generation to occur;*
- [2]

Multiple sclerosis is a disease in which the myelin sheath surrounding neurones are destroyed and so the neurones become de-myelinated. People with multiple sclerosis produce slow responses to stimuli.

(c) Explain how myelination of neurones helps in speeding up responses.

1. myelin sheath contains lipids acting as electrical insulator agst depolarization;

2. Na^+ & K^+ cannot pass through axon memb;

3. local circuit produced by A.P. at 1 node of Ranvier jumps to next node causing another A.P. to be propagated;

4. via saltatory conduction;

[2]

The black widow spider venom contains a toxin called latrotoxin that affects the normal functioning of the synapse. The bite of the black widow spider initially produces an increase in neuromuscular activity that leads to painful skeletal muscle spasms and rigidity. This phase of hyperexcitability is rapidly followed by progressive failure of neuromuscular transmission and paralysis. Experiments have found that latrotoxin can form permeable calcium ion pores on the presynaptic membrane.

(d) Suggest what happens at the synapse of a person bitten by a black widow spider.

1. as A.P. reaches post synaptic memb @ pre-synaptic knob, more voltage gated Ca^{2+} channels open due to depolarization;

2. causes greater influx of Ca^{2+} into synaptic knob;

3. results in more synaptic vesicles containing neurotransmitters to translocate towards pre-synaptic memb;

4. fusion of synaptic vesicles with post-synaptic memb causing release of more neurotransmitters into synaptic cleft via exocytosis;

[2]

[Total: 12]

- 7 Finches inhabiting the Galapagos islands are thought to have evolved from a population of a single species which managed to reach the islands thousands of years ago from South America. A phylogenetic tree was constructed by comparing base sequences in the DNA. Fig. 7.1 shows the evolutionary relationship.

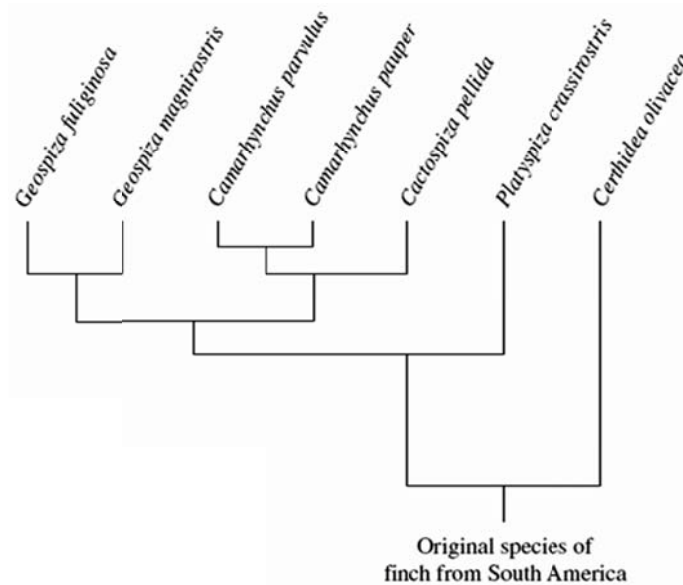


Fig. 7.1

- (a) With reference to Fig. 7.1, state
- (i) which present-day species of finch evolved first on the Galapagos islands;
Certhidea olivacea ® if not underlined [1]
- (ii) the number of genera that are present.
5 [1]
- (iii) the two most closely related genera.
Camarhynchus* & *Cactospiza [1]
- (b)(i) State **one** other type of evidence that can be used to determine evolutionary relationships.
anatomical homology / embryological homology / fossil (paleontology);
 ® ***homology (too vague)***
 ® ***molecular unless specify aa seq*** [1]
- (ii) State **one** advantage of using DNA base sequences to determine evolutionary relationships.
quantitative (measureable) / objective / AVP ® ***accurate (too vague)*** [1]
- (iii) Explain why other forms of evidence are needed to determine evolutionary relationships despite the advantage of using DNA sequences.
1. may not find sufficient homologous regions / gene loci for cf;
2. DNA diff may not lead to speciation;
3. lack of DNA in extinct spp / fossils;
 ® ***cost / req specialized techniques*** [2]

Two species of finches from the *Geospiza* genus living on different islands in the

Galapagos was observed to have different beak shapes. Both species are seed eaters. Fig. 7.2 shows the beak morphology of the two species.



Geospiza fuliginosa

Geospiza magnirostris

Fig. 7.2

- (c) With reference to Darwin's theory of natural selection, explain the difference in beak morphology of the two finch species.

1. ✓ of variation in beak size & depth / phenotype in ancestral popn;
2. finches occupy diff niches / habitats on Galapagos islands with diff selection pressure;
3. diff seed type / texture / hardness / location / accessibility ® predation;
4. niche with small seeds / narrow location, narrow beak type is selected for;
5. niche with large / hard seeds have greater access to food i.e. broad beak type is selected for;
6. finches with favourable / adaptable phenotype experience higher survival rates, greater repro success, passing down alleles coding for phenotype to offspring resulting in Δ in allelic freq coding for beaks in the 2 popns' gene pools & thus diff beak phenotype btw G. fuliginosa & G. magnirostris;
7. geographical isolation btw islands reduce (not prevent) interbreeding thus gene flow, ting genetic diversity btw the 2 popns leading to speciation;

[3]

[Total: 10]

Section B

- 8 (a) Compare and contrast the ways in which lambda phage and human immunodeficiency virus (HIV) reproduce themselves. [8]

Similarities:

1. Both attach to their specific host cells at receptor sites on host cell surface memb;
2. Both are obligate intracellular parasites/make use of their host cell's resources for synthesis of viral prots & nucleic acids;
3. Both integrate their viral DNA into host cell genome/viral DNA replicates as part of host's DNA every time cell divides;

Differences:

Feature	λ phage	HIV
Penetration / entry	4. Does not undergo fusion with host CSM;	5. HIV env fuses with host CSM;
	6. Lowers base plate on cell wall & injects only its d/s DNA;	7. Releases its nucleocapsid (s/s RNA & reverse transcriptase) into cytoplasm;
Uncoating	8. No uncoating req'd as capsid does not enter host cell;	9. Uncoating of nucleocapsid to release genome & enz;
Fate of viral nucleic acids	10. No RT as viral nucleic acid is d/s DNA, used directly as template for syn of viral prots & nucleic acid or integrated into host chr;	11. s/s RNA converted to d/s DNA using RT;
	12. Prophage excised from host cell's chr upon spontaneous induction;	13. Provirus is not excised as viral mRNA is transcribed from viral DNA together with host cell's genes;
Synthesis of viral nucleic acids / replication of viral genome	14. d/s DNA is syn directly using host cell's enz e.g. DNA pol;	15. s/s DNA is syn indirectly using RT & host cell's enz such as RNA pol;
Integration of viral glycoprots into cell memb	16. No integration of viral glycoprots into host's cell memb;	17. Integration of viral glycoprots (gp 120 & gp 41) into host cell memb;
Site of syn of viral nucleic acids / replication of viral genome	18. Cytoplasm of bacterium;	19. Cytoplasm & nucleus of human cell i.e. T helper cell;
Release/Exit	20. Host cell is lysed to release new viruses;	21. New viruses bud off from host CSM;

- (b) With reference to a bacteriophage and its reproductive cycle, outline how generalized transduction and transformation could result in a gene transfer between bacteria. [8]

Generalized transduction

1. ***Use of T4 bacteriophages to carry bacterial genes from one host to another;***
2. ***Adsorption stage, attachment of tail fibre to host cell;***
3. ***Penetration stage, tail sheath contracts to inject viral DNA into host cell;***
4. ***Viral DNA undergoes replication using host cell DNA pol to produce multiple copies of itself;***
5. ***Host cell DNA also degraded by phage coded enzyme;***
6. ***Viral DNA undergoes protein synthesis using host cell RNA pol & ribosomes to produce viral mRNA & viral proteins;***
7. ***Viral DNA & viral proteins packaged into capsid;***
8. ***Also erroneously package degraded bacterium DNA into some capsids of new viral phages;***
9. ***Host cell lysed and new phages released;***
10. ***New phage (containing degraded DNA of previous bacterium) infects recipient bacterium by injecting bacterial DNA;***
11. ***Homologous recombination may occur, newly introduced bacterial genes replace homologous genes on recipient bacterium genome;***
12. ***Result in transfer of donor genes into recipient bacterium***

Transformation

13. ***Uptake of exogenous DNA into competent bacterium;***
14. ***Exogenous DNA originate from previous bacterial cells that has undergone lysis;***
15. ***DNA degraded by enz;***
16. ***DNA introduced into new bacterium undergoes homologous recombination;***
17. ***Cause exchange of alleles in new bacterium;***

- (c) Some bacteriophages undergo both lysogenic and lytic cycles.

Suggest reasons why it may be advantageous for a bacteriophage to have a lysogenic cycle. [4]

1. ***Under unfavourable conditions (e.g. deprived of nutrients), allow phage to be dormant in host cell;***

2. *Does not undergo replication & reproduction, thus saves energy & resources;*
3. *Phages exposed to environmental hazards for months or years;*
4. *Bacteria able to survive & carry virus genome;*
5. *Synthesise new copies as they reproduce;*
6. *Survival is ensured since host cell survives;*
7. *Viral genome is replicated each time host cell replicates, allowing a large no. of viral particles to be assembled upon spontaneous induction;*
8. *Resulting in full blown infection/increase efficiency of viral infection;*
9. *Allow virus to survive when there is an effective anti-viral drug present;*

[Total: 20]

9 (a) Outline the main stages in Krebs cycle.

[6]

1. *occurs in mitochondrial matrix;*
2. *acetyl CoA formed fr pyruvate in link rxn;*
3. *combines with OAA (4C) → citrate (6C);*
4. *citrate undergo oxidative decarboxylation to 5C acid involving;*
5. *removal of CO₂ & reduction of NAD to NADH;*
6. *5C undergo oxidative decarboxylation to 5C acid involving;*
7. *removal of CO₂ & reduction of NAD to NADH;*
8. *4C undergo series of rxns to regenerate OAA;*
9. *producing 1 ATP / GTP via substrate level phosphorylation;*
10. *FAD → FADH;*
11. *NAD → NADH;*
12. *NADH & FADH carries H & e-s to ETC for OP;*

(b) Explain the significance of oxidative phosphorylation in aerobic respiration.

[8]

1. *occurs in at the inner mitochondrial memb / cristae;*
2. *in presence of O₂ / during aerobic resp;*
3. *produces 34 out of 38 ATP mols fr oxidation of 1 glc mol;*
4. *NADH & FADH oxidised at beginning of ETC, to NAD & FAD respectively;*
5. *regenerated NAD & FAD goes back to glycolysis, link rxn & Krebs cycle;*
6. *allow continuous flow of glycolysis, link rxn & Krebs cycle;*
7. *high energy e- trspted along ETC thro repeated [O]-[R] rxns;*
8. *down energy level by e- carriers;*
9. *en lost harnessed / used by e- carriers to pump H⁺ across inner mito mem fr mito matrix to intermembranal space;*

10. *establish a proton grad;*
 11. *H⁺ diffuse back into matrix down conc. grad;*
 12. *thro' F₀ subunit of stalked particle;*
 13. *proton motive force rotates the F₁ subunit of stalked particle;*
 14. *drives phosphorylation of ADP by ATP synthase;*
 15. *allows ATP synthesis via chemiosmosis;*
 16. *O₂ acts as final e⁻ acceptor, reduced to H₂O catalysed by cytochrome oxidase;*
 17. *1 NADH mol can give rise to 3 mols of ATP; (accept 2.5)*
 18. *1 FADH mol can give rise to 2 mols of ATP; (accept 1.5)*
- (c) With reference to the term 'limiting factor', discuss how changes in light intensity affects the rate of photosynthesis. [6]
1. *limiting factor refers to the variable / factor in the rate-limiting step;*
 2. *usually the factor which is in shortest supply;*
 3. *light is needed for photoactivation of chl a rxn ctr in light dep stage of PS;*
 4. *LI is a limiting factor of PS @ low intensity; ® initial stages*
 5. *rate of PS ↑es linearly / directly proportionally with ↑ing LI;*
 6. *↑er LI => higher rate of photoactivation of chl a in PSI & PSII => ↑er PS rate;*
 7. *light compensation pt where PS rate = resp rate;*
 8. *@ high LI, rate of PS starts to ↑ at ↓ing rate / approach plateau;*
 9. *further ↑ in LI => no further ↑ in PS rate / reached max;*
 10. *light saturation pt being reached;*
 11. *LI no longer limiting factor;*
 12. *other factors become limiting factor e.g. CO₂ conc / temp*

[Total: 20]